Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/03/2006
PI Name:	Faghri, Mohammad Ph.D.		
Project Title:	Phase Change in Low and Jittering Gravity Environment Simulated via Electromagnetic Field		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02881-0805	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Faghri, M.		
COI Name (Institution):	Charmchi, M. (Univ. of Massachusetts-Lowell) Kalluri, D. K. M. (University of Massachusetts-Lowell)		
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Performance Goal No.:			
Performance Goal Text:			
	Transport processes associated with melting of an electrically conducting Phase Change Material (PCM), placed inside a rectangular enclosure, under low-gravity environment, and in the presence of an electromagnetic field is simulated numerically and experimentally. Electromagnetic forces damp the natural convection as well as the flow induced by sedimentation and/or floatation, and thereby simulating the low gravity environment of outer space. Computational experiments are conducted for both sidewall heating and top-wall heating in both two- and three-dimensional rectangular enclosures. The governing equations are discretized using a control-volume-based finite difference scheme. Numerical solutions are obtained for true low-gravity environment as well as for the simulated-low-gravity conditions. The effects of electromagnetic field on the natural convection, solid phase floatation/sedimentation, liquid-solid interface location, solid melting rate, and flow patterns are investigated. It is found that the melting under low-gravity environment can reasonably be simulated on earth via applying a strong horizontal magnetic field. However, the flow		

Task Description:	patterns obtained for the true low-gravity cases are not similar to the corresponding cases solved for the simulated-low-gravity environment.On the other hand, it is shown that under electormagnetic field simulation, where the Lorentz force is caused by the transverse electric and magnetic fields, it is possible to minimize the flow field distortiion caused by the high magnetic field and therefore achieving a much better simulation of low-gravity. Both solidification and melting experiments were performed. These set of experiments are for sidewall phase change processes and therefore, addresses the fixed solid phase conditions. The effects of magnetic field on phase change rate and on the shape of the solid/melt interface are studied. The solid thickness is measured via ultrasonic techniques and the solid/melt interface is mapped using florescent light shadowgraphy through a transparent window. The collected data consist of temperature history, ultrasonic detection of the interface, florescent light shadowgraphy and solid phase volume fraction. The results showed that the presence of magnetic field had a marked effect on melting and natural convection whereas; phase change convection was noticeable in the solidification cases. The objective of the experimental part of the project is to design the Experimental Apparatus and Measurement System. Experiments will be conducted to observe solidification and melting processes under low gravity condditions induced by electromagnetic fields and to measure the temperature field and the solid thickness as a function of time and wall temperature conditions. The objective of the numerical part of the project is to model and solve the phase change problem under low gravity environment in two- and three-dimensional enclosure. Also, preliminary computations will be conducted to assist with the design of the experiment. Tasks Performed in the Numerical Part A (M. Faghri)Development of the computer codeValidating the code in the limiting casesFormulation of the ProblemSimulation of some cases to assist experiments. Also conduct computational experiments for both fixed(sidewall heating)and unfixed(top-wall heating)PCM in three-dimensional enclsoures. Tasks Performed in the Experiment Part B (M. Charmchi)Review of experiment design requirements: test cell dimensions, materials selections, boundary conditions, fabrication procedures, and assembly considerations.Instrumentation Requirements: data requirements and data acquisitions:Temperature measurements (sensors specifications and positions)Thermal environmental control (power requirements -heating and cooling- and equipment specifications)Data acquisition system (I/O boards, PC interface, and DAQ software). Conduct experiments to observe solidification and melting processes under low gravity conditions simulated by electromagnetic fields and measure temperature field and solid thickness as a funciton of time and the wall temperature conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Task progress in numerical part of the project: The application of electromagnetic fields both oscillating and static have been utilized to enhance melting of electrically conducting phase change material (gallium) under a low gravity environment. Both transverse electric and oscillating or static magnetic fields are used to generate an oscillating Lorentz force to enhance convective mechanisms loss when the level of gravity decreases. Furthermore, the influence of surface tension has been studied to determine its local and global effects on the melting system. The problem is formulated as a one-domain by employing an enthalpy-based transformation of the energy equation, which allows for one set of governing equations to be solved. The governing equations are then discretized using a control-volume-based finite difference scheme. The results show that the application of either static or oscillating electromagnetic fields can be used to enhance the rate of melting of the phase change material over the melt rate seen under true low gravity alone. In addition it has been determined that a static electromagnetic field can be used to enhance the melting rate more than that application of an oscillating electromagnetic field. When the effect of surface tension is introduced increased melting could be achieved for certain combinations of field strength and type (static vs. oscillating). Finally, the strongly three dimensional and unique flow fields caused by the application of either static or oscillating electromagnetic fields have been documented. Task progress in experimental part of the project and validation with numerical results: An experimental and numerical validation is conducted on melting behavior of a pure metal in the presence of a static magnetic field. When a transverse magnetic field is present, the resulting Lorentz forces will dampen the convective flows. Buoyancy driven flow is the focus of the study. Hartmann number, a dimensionless parameter proportional to the strength of magnetic field, dominates the convection flow suppression. The effects of the magnetic strength on melting rate and on the profile of solid/melt interface are studied. The experiments are conducted with pure Gallium as a phase change material inside a rectangular test cell. The solid/melt interface at the side center position is measured by an ultrasound device and its profile is mapped via the florescent light shadowgraphy. Temperature measurements and volume expansion/contraction tracking are used to verify the experimental result. The results show that the magnetic strength has a pronounced effect on the melting rate and the interface profile. The numerical simulation agrees well with the experimental data especially at larger Hartmann numbers.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Veilleux DL, Gonvalves E, Faghri M, Asako Y, Charmchi M. "Phase change in a three-dimensional rectangular cavity under electromagnetically simulated low gravity: top wall heating with an unfixed material." Numerical Heat Transfer, Part A: Applications: An International Journal of Computation and Methodology, 2005 Dec;48(9):849–78. http://dx.doi.org/10.1080/10407780591006903 48:849-878, 2005. , Dec-2005
Articles in Peer-reviewed Journals	Veilleux DL, Gonvalves E, Faghri M, Asako Y, Charmchi M. "Phase change in a three-dimensional rectangular cavity under electromagnetically simulated low-gravity: Side wall heating." International Journal of Numerical Methods for Heat and Fluid Flow. 2005;15(7):710-39. http://dx.doi.org/10.1108/09615530510613898 , Dec-2005
Papers from Meeting Proceedings	Zhang H, Sun H, Charmchi M, Veilleux D, Faghri M. "Experimental and Numerical Investigation of Melting in the Presence of a Magnetic Field--Simulation of Low Gravity Environment." 2005 ASME International Mechanical Engineering Congress and Exposition, Orlando, Fl, November 5-11, 2005. 2005 ASME International Mechanical Engineering Congress and Exposition, Orlando, Fl, November 5-11, 2005. , Nov-2005